

Bilkent University EEE202 LAB #1

First-Order RC Circuit's Response to Rectangular and Sinusoidal Pulses

Preliminary Work

Consider the RC circuit shown in Figure 1. The values of R and C are chosen such that the time constant

$$\tau = RC$$

lies in the range $0.5 \text{ ms} < \tau < 10 \text{ ms}$.

The values of R and C are assigned and should be checked from the Lab 1 RC Assignment file.

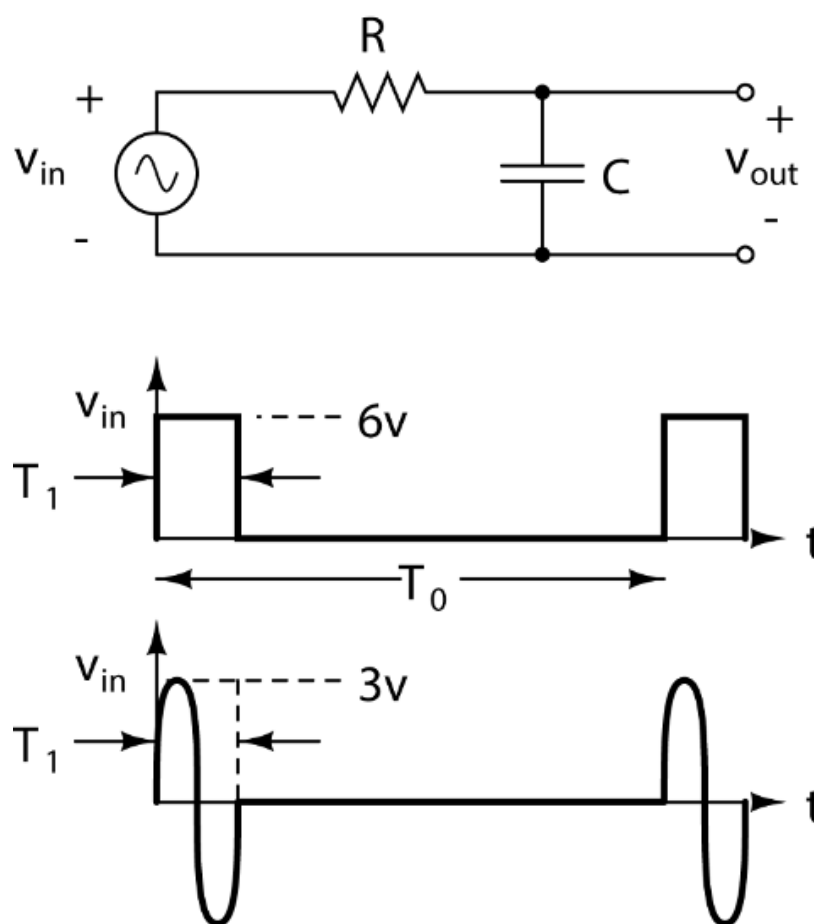


Figure 1: First-order RC circuit used in the experiment.

1. Preliminary Part 1: Rectangular Pulse Response

Calculate the response of the RC circuit to a rectangular pulse of magnitude 6 V with duration T_1 and repetition period T_0 .

Choose:

$$\frac{\tau}{2} < T_1 < 3\tau, \quad 30\tau < T_0 < 50\tau$$

so that the capacitor voltage is zero at the beginning of each repetition period.

Derive two analytical expressions for the capacitor voltage:

- for $0 < t < T_1$
- for $T_1 < t < T_0$

Use the capacitor voltage at $t = T_1$ as the initial condition for the second time interval. Plot the expected waveform using **MATLAB**, and simulate the circuit using **LTSpice**. Compare the analytical solution with the simulation results, and include the comparison in the report.

Note: All expressions must be derived in the time domain. Phasor analysis will not be accepted for this part.

2. Preliminary Part 2: Sinusoidal Pulse Response

Calculate the response of the RC circuit to a single sinusoidal pulse with peak value 3 V and period T_1 , repeated every T_0 .

Choose:

$$\frac{\tau}{2} < T_1 < 3\tau, \quad 30\tau < T_0 < 50\tau$$

Since $T_0 \gg \tau$, assume the initial capacitor voltage is zero. Derive analytical expressions for:

- $0 < t < T_1$
- $T_1 < t < T_0$

Use the capacitor voltage at $t = T_1$ as the initial condition for the second interval. Plot the expected waveform using **MATLAB**, and simulate the circuit using **LTSpice**. Compare the analytical solution with the simulation results, and include the comparison in the report.

Note: All expressions must be derived in the time domain. Phasor analysis will not be accepted for this part.

Experimental Work

Measurement Setup

- Connect the **TRIG OUTPUT** at the back of the signal generator to the **EXT TRIG** input of the oscilloscope using a coaxial cable.
- Set the oscilloscope trigger settings as follows:
 - Trigger Source: **EXT**
 - Trigger Type: **Edge**
 - Trigger Slope: **Rising**
 - Trigger Mode: **Normal**
 - Trigger Coupling: **DC**
- Set the horizontal time scale to approximately one time constant (τ) per division.

Rectangular Pulse Excitation

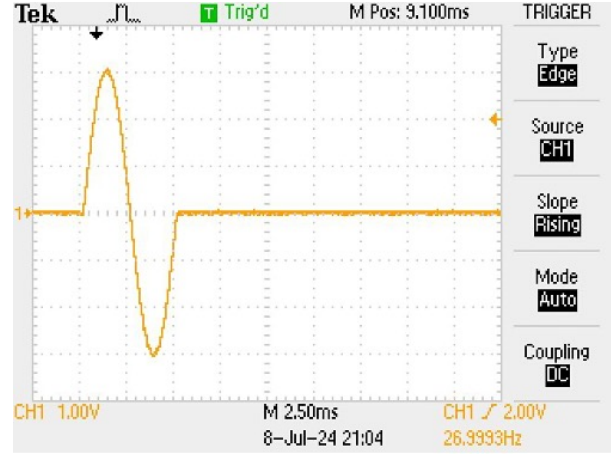
To configure the signal generator (DS345) to generate a rectangular pulse of 6 V with pulse width T_1 and repetition period T_0 , follow the steps below:

1. Press [SHIFT] [DEFAULTS] to reset the generator to its default settings.
2. Press [FREQ] and set the frequency to $1/(2T_1)$.
3. Press [AMPL] and set the amplitude to $3V_{pp}$.
4. Press [OFFST] and set the DC offset to 1.5 V to shift the baseline to zero volts.
5. Press [PHASE] and set the phase to 180° to obtain a positive pulse.
6. Select the **Square Wave** function using the down arrow in the **Function** menu.
7. Press the bottom arrow in the **SWEEP/MODULATE** menu to enter **BURST** mode.
8. Press [SHIFT] [TRIG SOURCE] and use the left arrow to select **Internal RATE** mode.
9. Press [SHIFT] [TRIG RATE] and set the trigger rate to $1/T_0$.
10. Press [SHIFT] [BURST CNT] and set the burst count to **1**, so that a single pulse is generated every T_0 .
11. Press [SWEEP ON/OFF] to enable burst mode. The orange LED should turn on, and the green **TRIG'D** LED above the [FREQ] button should flicker at the repetition rate.

A sample generated input waveform is illustrated in Figure 2a.



(a) Rectangular pulse input signal



(b) Single-cycle sinusoidal pulse input signal

Figure 2: Input excitation signals applied to the RC circuit.

Sinusoidal Pulse Excitation

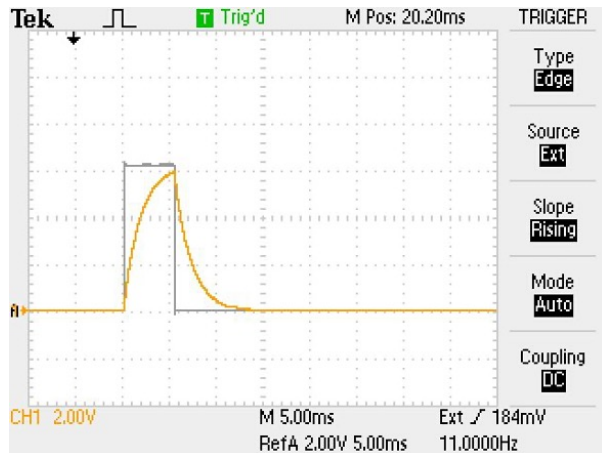
To generate a single-cycle sinusoidal pulse repeated every T_0 , configure the signal generator as follows:

1. Press [SHIFT] [DEFAULTS] to reset the generator.
2. Press [FREQ] and set the frequency to $1/T_1$.
3. Press [AMPL] and set the amplitude to $3V_{pp}$.
4. Press the bottom arrow in the **SWEEP/MODULATE** menu to enter **BURST** mode.
5. Press [SHIFT] [TRIG SOURCE] and select **Internal RATE** mode.
6. Press [SHIFT] [TRIG RATE] and set the trigger rate to $1/T_0$.
7. Press [SHIFT] [BURST CNT] and set the burst count to **1**, so that only one cycle of the sine wave is generated per period.
8. Press [SWEEP ON/OFF] to enable burst mode.

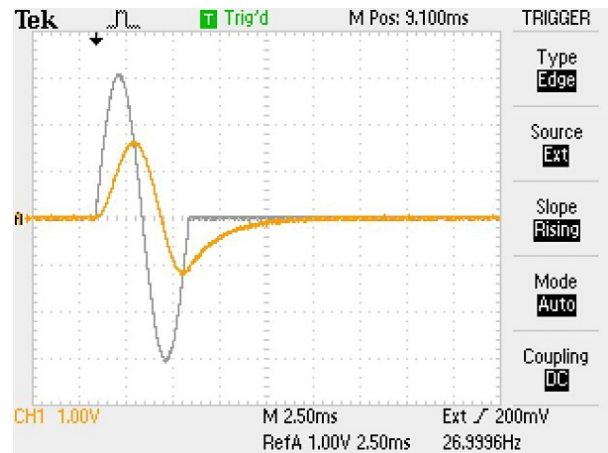
A sample generated sinusoidal pulse is shown in Figure 2b.

Simultaneous Measurement of Input and Output

1. Connect the oscilloscope probe to the input of the RC circuit.
2. Connect the second oscilloscope probe to the output of the RC circuit, across the capacitor.
3. Adjust the vertical scale and enable waveform averaging if the output signal is small compared to the input.



(a) Sample output response to rectangular pulse excitation



(b) Sample output response to sinusoidal pulse excitation

Figure 3: Measured output responses of the RC circuit.

Compare the experimentally measured waveforms with the analytical results obtained in the preliminary work. **Calculate the error rate and ensure that it is within the acceptable range of $\pm 20\%$. Then, present the outputs of your hardware outputs, simulation results and MATLAB calculations, including the error rate evaluation, to your TA in order to receive the check.**